Class activity

- Each group must bring 5 recent refereed articles saved in Zotero.
- Prepare a literature matrix with Author/Year, Problem, Method, Findings, Limitation, and Relevance.
- Identify which article supports the problem, method, technology, gap, and proof of concept.
- Write one synthesized paragraph using at least 3 sources.
- Check if the citations and reference entries match APA 7.

Create a Literature Matrix

This helps students move from collecting articles to comparing and synthesizing them.

Recommended columns

Author/Year	Problem	Method	Data	Major Finding	Limitation	Relevance to Study
Santos et al. (2024)	Demand forecasting	XGBoost	Historical sales data	Improved demand prediction	Limited to historical sales patterns	Supports the forecasting component
Cruz & Lim (2025)	Stockout prediction	Random Forest	Inventory and sales data	Identified products at risk of stockout	Did not include automated recommendations	Supports stockout analysis and risk indicators
Reyes et al. (2023)	Inventory monitoring	Monitoring system	Inventory records	Improved tracking of stock levels	Focused only on monitoring	Shows the gap in monitoring-only systems

Tip: The matrix shows patterns, common limitations, and possible research gaps.

Author/Year	Problem	Method	Data	Major Finding	Limitation	Relevance to Study
Li et al. (2025)	Crowded vessel-to-AIS matching ambiguities	YOLO-OBB and JVC global assignment algorithm for dark ship isolation	Bohai Sea optical, SAR, and AIS datasets	91% accuracy vessel detection	Small-vessel vulnerabilities and sensor inaccuracies	Supports AIS-SAR dark-vessel detection existence
(Hanmante et al., 2026)	Sensory asymmetry in dark-vessel surveillance	Multi-tiered architectural review of pixel-, feature-, and decision-level sensor fusion	Sentinel-1 SAR, Gaofen optical, and coastal AIS telemetry	Cuts false alarms 35–60%; pairwise accuracy drops below 75% in dense ports	Revisit latency; pairwise matching lacks relational graph modeling	2026 SOTA anchor; justifies spatiotemporal graph modeling for congested port waters
Zhang et al. (2023)	Ignored dynamic ship-to-ship interactions in dense traffic	Three-branch G-STGAN parallel architecture for spatiotemporal feature extraction	Three real-world AIS datasets representing varied traffic scenes	9.47% MAE improvement against SOTA (LSTM, GRU) baselines	Dependence on clean, uninterrupted AIS data	Provides architectural blueprint for trajectory reasoning and cross-modal graph fusion
(Xue et al., 2026)	Feature redundancy and inefficient fusion in multi-modal models.	Asymmetric multi-modal fusion network (AMFNet) using AIS density priors.	Combined synthetic aperture radar (SAR) imagery and AIS spatial datasets.	Outperforms symmetric baselines in ship detection efficiency.	Performance drops during low-contrast radar anomalies or missing AIS priors.	Supports advanced feature-level sensor fusion integrating AIS with remote sensing.
(do Nascimento et al., 2024)	Identifying illegal maritime activities through complex behaviors.	Hybrid framework combining data-driven analytics and expert rules fusion.	Real-world multi source maritime surveillance.	Robust identification of illegal operational patterns.	Rigid reliance on predefined rule boundaries.	Supports problem identification for illegal maritime surveillance.

Study 1 - problem
Study 2 - Gap

Study 3 - POC
Study 4 - Technology
Ano focus ng synthesis dark-vessel candidate?not sure pa if anong theme
Synthesis

References

Hanmante, S., Kolhar, S., & Patil, S. (2026). A comprehensive review of multimodal artificial intelligence techniques in maritime surveillance using AIS, SAR, and optical data fusion. *Array*, *30*, 100931. https://doi.org/10.1016/j.array.2026.100931

Li, F., Yu, K., Yuan, C., Tian, Y., Yang, G., Yin, K., & Li, Y. (2025). Dark Ship Detection via Optical and SAR Collaboration: An Improved Multi-Feature Association Method Between Remote Sensing Images and AIS Data. *Remote Sensing*, *17*(13), 2201. https://doi.org/10.3390/rs17132201

Zhang, X., Liu, J., Gong, P., Chen, C., Han, B., & Wu, Z. (2023). Trajectory prediction of seagoing ships in dynamic traffic scenes via a gated spatio-temporal graph aggregation network. *Ocean Engineering*, *287*, 115886. https://doi.org/10.1016/j.oceaneng.2023.115886